

a brain tumor. Therefore, he reports almost no emotional experiences. He shows no impatience, no frustration; he experiences no joy from music or art; he rarely expresses anger; he describes his brain surgery and the resulting deterioration of his life with calm detachment, as if he were describing events that happened to a stranger. Besides his loss of emotions, he has great difficulty making or following any reasonable plans. If given an array of information, he can discuss the probable outcome of each decision he might make, but after describing those outcomes, he seems to have no idea which decision to make. Or if he does announce a decision, he abandons it soon after. As a result, Elliot cannot keep a job, cannot invest his money intelligently, and cannot maintain normal relationships with his friends or acquaintances.

According to Damasio (1994), Elliot's difficulty with decision-making is closely related to his lack of emotions. Ordinarily, when you or I consider possible decisions, we contemplate the possible outcomes and feel a brief "as if" emotion with each. For example, you consider a job offer from a company that pollutes the environment, and you get a feeling of revulsion, so you decide to reject the offer. When you consider another offer from a company that recently fired your best friend, you imagine the unpleasant scene when you next face that friend, and again you decide to reject the offer. You continue until you come to a job offer that will make you feel good if you accept it. Now, imagine eliminating all those emotions, and you can see that the capacity for emotion is helpful, even necessary, for making many decisions.

Many psychologists therefore speak of emotional intelligence, the ability to perceive, imagine, and understand emotions and to use that information in decision-making (Mayer & Salovey, 1995, 1997). For example, imagine that you are standing in line for lunch but you have left a bit of a gap between you and the person in front. Someone else steps into the line in front of you. You have to judge whether this person was being rude, or merely making an honest mistake by not realizing you were in this line. Should you scream threats and insults? Or should you calmly inform this person of the mistake? Or ignore it altogether?

Or imagine a man who is trying to decide whether to tell a certain joke to a couple of women. He needs to gauge how they will probably respond: Will they enjoy it, or will they regard it as sexist, offensive, or worse? Suppose he tells it. Now, the women have to judge his intentions: Did he mean no harm, or was this joke potentially the start of some sort of sexual harassment?

In short, we frequently need to make complicated judgments, and people who often make the wrong judgments can get themselves into trouble. Undoubtedly, some people are better than others at judging the emotional content of a social situation, although psychologists know little about how to measure this sort of skill. Ultimately, it would be useful to find ways to teach emotional

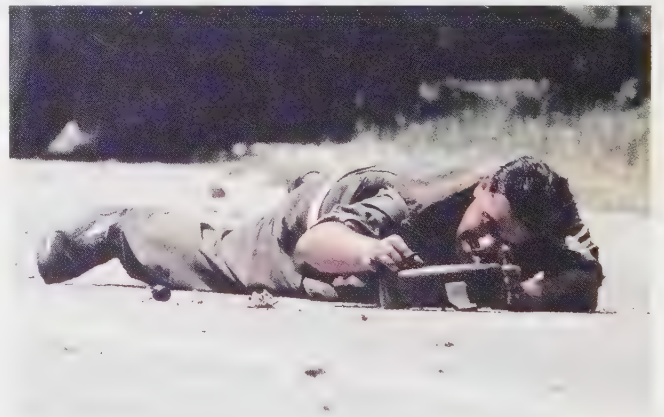
intelligence, although no one is yet sure how. As with any other kind of teaching, it is possible to teach with good intentions and still produce more confusion than clarity (Mayer & Salovey, 1997).

Excitement and Physiological Arousal

Originally, the word *emotion* was a general term for any sort of turbulent movement. People used to talk about thunder as an "emotion of the atmosphere." Eventually, the word came to refer only to feelings associated with vigorous motion of the body, such as fear, anger, and joy.

We experience emotional arousal, or excitement, when we have a strong tendency either to approach or to avoid something, generally in an energetic way. For example, love includes a strong drive to be close to another person. Anger includes a tendency to charge toward someone and to attack either verbally or physically. Fear and disgust are associated with a tendency to escape. All emotions share certain features related to physiological arousal. So, although anger, fear, and happiness are very different emotional states, we can express any one of these by screaming or by engaging in frenzied activity.

"But wait," you say. "Sometimes when I feel highly emotional, I can hardly do anything at all. Like the time I borrowed a friend's car and then wrecked it. When I had to explain what had happened to the car, I could hardly speak." True, but even then your emotion was associated with a tendency to take vigorous action. While you were reporting the wreck to your friend, you undoubtedly felt a strong urge to run away. Although you suppressed that urge, it made itself apparent in your trembling voice and shaking hands.



Ordinarily, an emotional state elicits a tendency toward vigorous action, even if we suppress that tendency. Here, a soldier disarms a mine during the war in the former Yugoslavia. No doubt he feels an intense desire to run away, and no doubt his heart is racing, but he manages to restrain himself from acting upon these impulses.

The Role of the Autonomic Nervous System

Any stimulus that arouses an emotion—such as a hug, a fire alarm, or a slap on the face—alters the activity of the **autonomic nervous system, the section of the nervous system that controls the functioning of the internal organs.** The word *autonomic* means independent; biologists once believed that the autonomic nervous system operated independently of the brain and the spinal cord. We now know that the brain and the spinal cord send messages to alter the activity of the autonomic nervous system, but we continue to use the term *autonomic*.

The autonomic nervous system consists of the sympathetic and the parasympathetic nervous systems (Figure 12.2). **Two chains of neuron clusters just to the left and right of the spinal cord make up the sympathetic nervous system, which arouses the body for “fight or flight.”** (See Figure 12.3.) For example, if someone charges at you, you may have to choose between fighting and running away,

but in either case, your sympathetic nervous system prepares you for a burst of vigorous activity. It does so by increasing your heart rate, breathing rate, production of sweat, and flow of epinephrine (EP-i-NEF-rin; also known as adrenaline).

The **parasympathetic nervous system** consists of neurons whose axons extend from the medulla (Figure 12.2) and the lower part of the spinal cord to neuron clusters near the internal organs. **The parasympathetic nervous system decreases the heart rate, promotes digestion, and in general supports nonemergency functions.** Both the sympathetic and parasympathetic systems send axons to various organs of the body, such as the heart and digestive organs. A few organs, such as the adrenal gland, receive input from only the sympathetic nervous system.

Both systems are constantly active. The alternation or shifting balance between the two systems helps to keep the body in homeostasis (see Chapter 10), but one system can be temporarily dominant over the other. An emergency that demands a vigorous response will predominantly activate



FIGURE 12.2 The autonomic nervous system consists of the sympathetic and parasympathetic nervous systems, which sometimes act in opposing ways and sometimes in cooperative ways. The sympathetic nervous system readies the body for emergency action; the parasympathetic nervous system supports digestive and other nonemergency actions.



FIGURE 12.3 The sympathetic nervous system prepares the body for a vigorous burst of activity.

the sympathetic system; a restful situation predominantly activates the parasympathetic system. Some situations activate parts of both systems at once (Berntson, Cacioppo, & Quigley, 1993). For example, a frightening situation may increase your heart rate and sweating (sympathetic responses) and also promote bowel and bladder evacuation (parasympathetic responses). Remember the last time you were seriously frightened. Chances are, your heart was beating wildly, you found yourself gasping for breath, and you were afraid you were going to lose your bladder control. Both your sympathetic and parasympathetic nervous systems were responsible for those responses.

The Opponent-Process Principle of Emotions

After the cessation of a stimulus that has excited sympathetic activity, the system “rebounds” with increased parasympathetic activity (Gellhorn, 1970). (See Figure 12.4.) For example, while you are running away from an attacker, your sympathetic nervous system increases your heart rate and your breathing rate. If the police suddenly in-

tercept your attacker, your sympathetic arousal ceases, and your parasympathetic system becomes highly activated on the rebound. If the rebound is large enough, a person who has just escaped from danger may faint because of the sudden decrease in heart rate.

This tendency is related to a larger, general principle, the **opponent-process principle of emotions** (Solomon, 1980; Solomon & Corbit, 1974). According to this principle, the removal of a stimulus that excites one emotion causes a swing to an opposite emotion (Figure 12.5). This principle is similar to the opponent-process principle of color vision (discussed in Chapter 4). Recall that, when you stare for a long time at one color and then look away, you see its opposite. (After staring at yellow, you see blue.) Solomon and Corbit suggest that the same principle holds true with emotional states.

For example, suppose you are making a parachute jump for the first time. As you start your fall, you probably experience a state akin to terror. As you continue to fall and your parachute opens, your terror begins to subside. When you land safely, your emotional state does not simply return to normal; it rebounds to relief. As time passes, your relief gradually fades until, at last, you return to a normal state. Figure 12.6 shows these changes in emotional response over time. Solomon and Corbit refer to the initial emotion as the A state and the opposite, rebound emotion as the B state.

Here is another example: You hear on the radio that you have just won \$1 million in the lottery; you become elated. Later, you discover that the winner was not you, but someone else with a similar name. Now, you feel sad, even though you “lost” something you never really had.

Solomon and Corbit further propose that repetition of an experience strengthens the B state but not the A state. For example, after you have made several parachute jumps, your rebound pleasure becomes greater and starts to occur earlier and earlier. Eventually, you may not be aware of any initial terror at all; the entire experience becomes pleasant.

Figure 12.6 illustrates the changes in emotional response that occur when the experience-and-rebound cycle is repeated many times. Note that the A state has become weaker and the B state has become stronger and more prolonged.

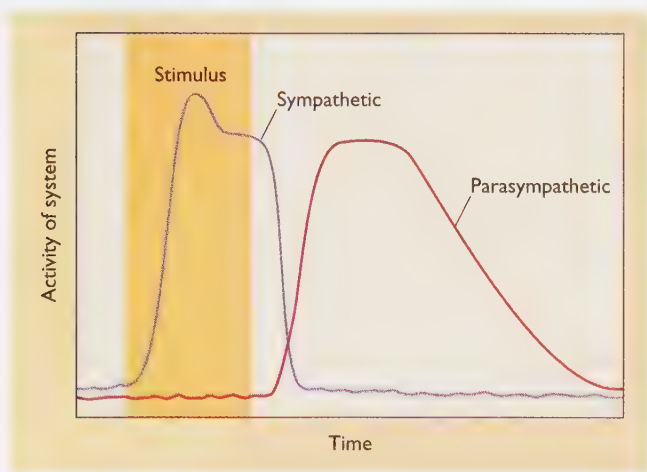


FIGURE 12.4 After removal of the stimulus eliciting the sympathetic response, that response is reduced, while the opposing parasympathetic response is enhanced. This is why people feel faint at the end of an exciting experience.



FIGURE 12.5 According to the *opponent-process principle of the emotions*, removing the stimulus for one emotion elicits a rebound to the opposite emotion. A hiker who sees a snake may feel terrified; when the threat passes, the terror gives way to relief and elation.

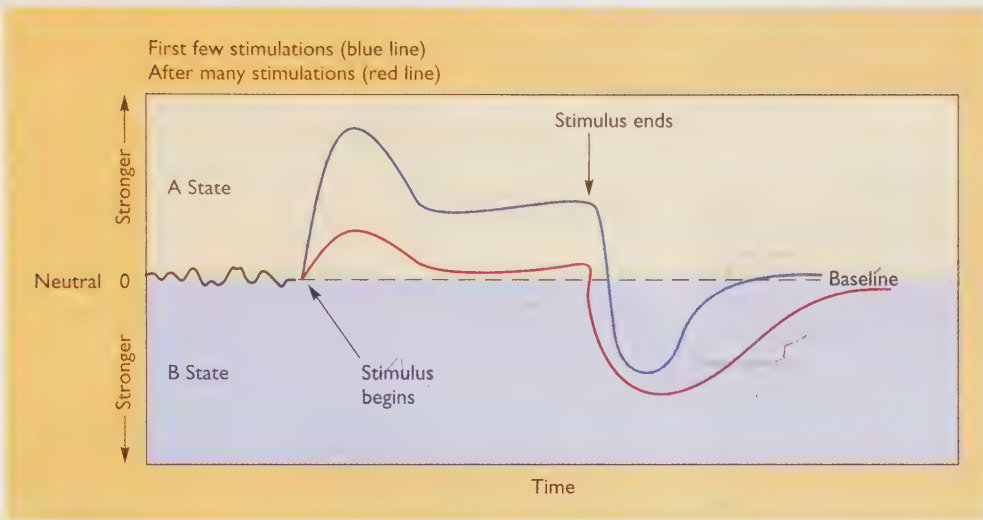


FIGURE 12.6 According to the *opponent-process theory of emotions*, removal of the stimulus for one emotion (A state) induces the opposite emotion (B state). The blue line shows emotional responses to a stimulus that is introduced and then withdrawn. The red line shows emotional responses to a stimulus that has been introduced and withdrawn repeatedly. Note how the intensity of the response changes over time. (Based on Solomon & Corbit, 1974.)

CONCEPT CHECKS

1. When you ride a roller coaster, does your heart rate increase or decrease? What happens after you get off the ride?
2. If we apply the opponent-process principle to the experiences that drugs produce, we can describe the initial “high” as the A state and the subsequent unpleasant withdrawal experience as the B state. If someone takes a drug repeatedly, how will the A state and the B state change? (Check your answers on page 452.)

The Sympathetic Nervous System and Lie Detection

Frequently, the police interview a suspect who denies any participation in a crime, or an employer interviews a prospective employee who claims to be skillful, honest, and trustworthy. In such cases, the interviewer would like to know whether the person is telling the truth. Unfortunately, most of us are not very good at detecting lies. In one study, a well-known British political commentator gave two consecutive interviews about his favorite films. In one, he told the truth; the other time, he lied on every question. Both interviews were broadcast on

radio and television and printed in a newspaper. Members of the public were invited to guess which of the interviews was the honest one and which one was full of lies. Of the more than 41,000 people who responded, only 53% got the right answer (Wiseman, 1995). Curiously, the television viewers were less accurate than the radio listeners or newspaper readers.

For centuries, people have tried to find a simple test to determine who is lying. One of the best-known attempts is the **polygraph**, an instrument that simultaneously records several indications of sympathetic nervous system arousal, generally including blood pressure, heart rate, breathing rate, and electrical conduction of the skin (Figure 12.7). (Slight sweating, a sympathetic nervous system response, increases electrical conduction of the skin.) The assumption is that people feel nervous when they lie, and consequently their sympathetic nervous system will show more arousal when they lie than when they tell the truth.

The polygraph sometimes accomplishes its goal simply because an accused person hooked up to a polygraph believes that it can detect lies and confesses, “Oh, what’s the use. You’re going to figure it out now anyway, so I may as well tell you . . .” But with people who do not break down and confess, how effectively does a polygraph detect lying? Let’s examine the evidence.

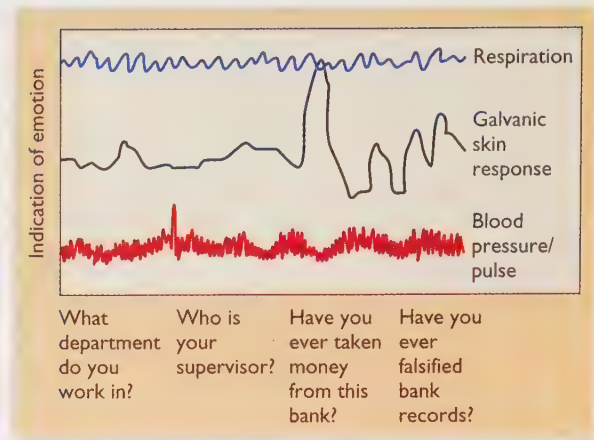


FIGURE 12.7 The polygraph, a method for detecting nervous arousal, is the basis for the so-called lie detector test. The polygraph operator asks a series of nonthreatening questions to establish baseline readings of the subject's autonomic responses, then asks questions relevant to an investigation. The underlying assumption is that an increase in arousal indicates nervousness, which in turn indicates lying. Unfortunately, a large percentage of innocent people also become nervous and therefore appear to be lying.

WHAT'S THE EVIDENCE? *The Effectiveness of a Polygraph in Detecting Lies*

Hypothesis Polygraph administrators will identify guilty suspects as liars more often than they identify innocent suspects as liars.

Method To test this hypothesis, the investigators need a sample of people who are known to be guilty and another sample who are known to be innocent but who are otherwise similar to the guilty people. In one study, the investigators selected 50 criminal cases where two suspects had taken a polygraph test, and one suspect had later confessed to the crime (Kleinmuntz & Szucko, 1984). Thus, the investigators knew which 50 suspects were guilty, and they knew that the 50 innocent people were similar enough to have been plausible suspects. It is important to note that all suspects had denied their guilt at the time of the polygraph test.

During the administration of the polygraph, suspects were asked two kinds of questions: *Relevant* questions pertained to the crime itself; for example, "Did you steal \$125 from the convenience store last Tuesday?" *Control* questions took the following form: "Have you ever taken anything of value that was not yours?" Theoretically, someone who robbed the convenience store should be more nervous about the first question; anyone else should be, if anything, more nervous about the second.

Six professional polygraph administrators examined all the polygraph results and judged which suspects were lying and which were telling the truth.

Results Figure 12.8 shows the results. The polygraph administrators did manage to identify 76% of the guilty suspects as liars; however, they also classified 37% of the innocent suspects as liars.

Interpretation Recall from the discussion of signal detection in Chapter 4 that, when a person is trying to determine whether something is present or absent, there are two possible correct decisions (green in the following diagram) and two possible errors (red).

	Stimulus actually present	Stimulus actually absent
Report stimulus present	Hit	False alarm
Report stimulus absent	Miss	Correct rejection

Polygraph administrators can also make two kinds of correct decisions and two kinds of errors:

	Actually lying	Actually telling truth
Polygraph says lying	Hit	False alarm
Polygraph says telling the truth	Miss	Correct rejection

A polygraph user obtains more "hits" than most people do on their own. However, most of the errors made by polygraph users are "false alarms" (also known as "false positives")—falsely identifying innocent people as lying. Given the usual belief that we would prefer to let guilty people go

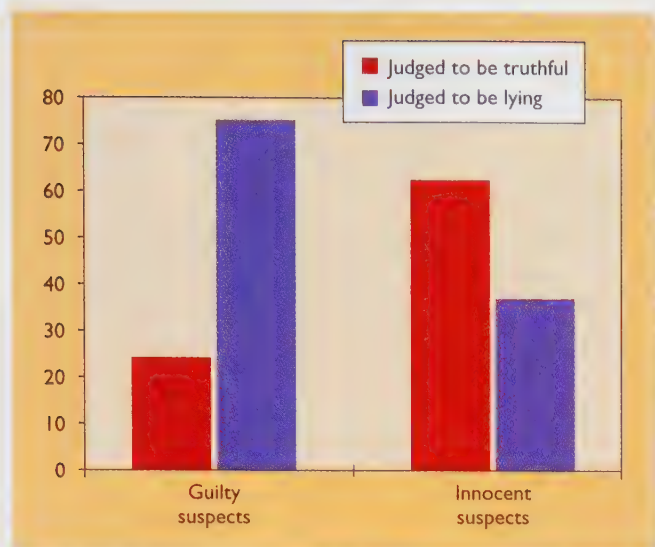


FIGURE 12.8 Polygraph examiners correctly identified 76% of guilty suspects as lying. However, they also identified 37% of innocent suspects as lying. (Based on data of Kleinmuntz & Szucko, 1984.)

free than to convict innocent people, these polygraph results are disturbing.

★ ★ ★

The results of this study are typical of similar research; in fact, this study may have even underestimated the tendency of polygraph administrators to call innocent people liars. In some studies, polygraph administrators have called half or more of the innocent people liars (Forman & McCauley, 1986; Horvath, 1977; Patrick & Iacono, 1989).

Imagine the consequences: Suppose we give polygraph tests to ten suspects, one of whom is guilty. The test would have about a 76% chance of identifying the guilty person as lying, but it would also identify several of the innocent people as liars. Or imagine giving a polygraph to all the people who work for a company, asking whether they have ever stolen from the company. Even if every employee is loyal and innocent, the test may identify almost half of them as liars. Because of the low accuracy of polygraph tests, the U.S. Congress passed a law in 1988 prohibiting private employers from giving polygraph tests to employees or job applicants, except under special circumstances (Camara, 1988). Polygraph results are only rarely admissible as evidence in a court of law.

An Alternative: The Guilty-Knowledge Test

The guilty-knowledge test, a modified version of the polygraph test, produces more accurate results by asking questions that should be threatening only to someone who knows the facts of a crime that have not been publicized (Lykken, 1979). Instead of asking, “Did you rob the gas station?” the interrogator asks, “Was the gas station robbed at

8:00 P.M.? at 10:30? at midnight? at 1:30 in the morning? Did the robber carry a gun? a knife? a club?” Someone who shows heightened arousal only in response to the correct details of the crime is presumed to have “guilty knowledge”—knowledge that only someone who had committed the crime or had talked to the guilty person would possess. The guilty-knowledge test, when properly administered, identifies a large percentage of guilty people and only rarely makes the mistake of classifying an innocent person as guilty (Balloun & Holmes, 1979; Bashore & Rapp, 1993; Elaad, 1990).

SOMETHING TO THINK ABOUT

How might the results of the guilty-knowledge test be biased by a questioner who knows the correct details of the crime? How should the test be administered to minimize that bias? ★

Pencil-and-Paper Integrity Tests

Suppose you are an employer who wants to know whether someone applying for a job at your company is likely to be an honest worker. Giving a polygraph test is illegal and would not be very accurate anyway, and you can’t give a guilty-knowledge test, because no one can have guilty knowledge about a crime that has not yet occurred. So what do you do?

One approach is to administer pencil-and-paper “integrity tests” that ask such questions as these:

- Have you ever stolen money or property from a previous employer?
- Do you think that most employees occasionally steal from their employers?
- On previous jobs, have you ever left work early while claiming to work a full day?
- Have you sometimes come to work while under the influence of illegal drugs?
- If you were sure you wouldn’t get caught, would you ever make personal long-distance phone calls and charge them to your employer?

You might imagine that anyone who has a history of dishonest dealings with previous employers would lie about it. Amazingly, many people fill out the questionnaire honestly, admitting a long history of past dishonesty. (Perhaps they assume the new employer will find out about this history anyway by checking with previous employers.) Research on such tests is limited, but it suggests that these tests manage to identify a good percentage of dishonest people (Camara & Schneider, 1994).

However, the integrity tests have two major problems: First, they misidentify some extremely ethical and scrupulous people who “confess” to being imperfect (Lilienfeld, Alliger, & Mitchell, 1995). For example, someone may read the question, “Have you ever stolen property from a previous employer?” and think, “Well, there was that one time

when I used a business envelope to mail a personal letter, and technically that was stealing."

Second, if the tests become widely used, job applicants will surely learn how to pretend to be honest: Say you have never stolen from an employer, you have never cheated your employer, you don't think most other people do either, and you think anyone who does should be punished. Test designers can then reword the questions to make their point less obvious, asking questions about conscientiousness, impulse control, and other characteristics that correlate with honesty. They can also include *social desirability* items: People who say no to items like *Occasionally I overeat* or who say yes to *I donate money to every worthy charity that I know* can be suspected of answering dishonestly to the more relevant items. In short, it may be possible to design valid integrity tests, but at present we have no guarantees (Alliger, Lilienfeld, & Mitchell, 1996).

Three General Theories of Emotion

Psychologists generally agree that emotions are related to the activity of the autonomic nervous system and the activity of the body in general. The nature of that relationship is less clear. Let us consider three theories that have offered different descriptions of that relationship.

The James-Lange Theory of Emotions

In 1884, William James and Carl Lange independently proposed a theory that immediately became highly influential and still remains highly influential, although psychologists have often misunderstood it. Indeed, even today people often confuse what James and Lange actually said, what they meant to say, and what casual readers thought they said (Ellsworth, 1994).

Common sense suggests that you feel sad and therefore you cry; you become afraid and therefore you tremble; you feel angry and therefore your heart beats faster and your face turns red. James and Lange turned this concept around. According to the **James-Lange theory** (Figure 12.9), a person's interpretation of a stimulus evokes the autonomic changes directly; the emotion is the perception of those changes. You decide that you are sad *because* you cry, you feel afraid *because* you tremble, and you feel angry *because* your heart is racing and your face is turning red. Similarly, the act of smiling brings happiness and frowning brings sadness. The body responses alone, James and Lange would agree, are not sufficient for emotions, but they are necessary. For example, trembling itself would not make you afraid. (You might be trembling because you are cold.) Nevertheless, if you did not tremble, you could not have the full experience of fear.

According to this theory, how would you know which emotion you are feeling? Actually, William James was skeptical of separating emotion into different categories such as

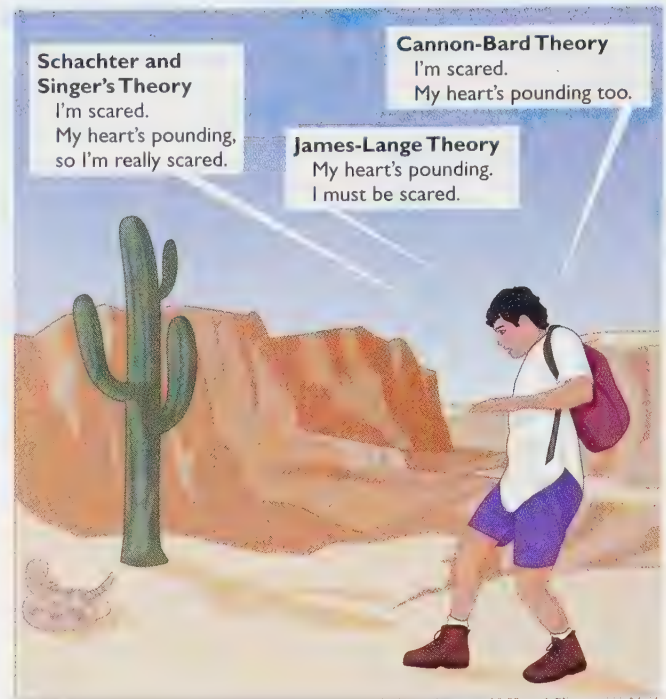


FIGURE 12.9 Three traditional theories of emotion differ concerning the relationship between physiological arousal and the cognitive experience of emotion. According to the James-Lange theory, the physiological arousal determines the nature of the emotion. According to the Cannon-Bard theory, the physiological arousal is independent of the cognitive experience. According to Schachter and Singer's theory, the physiological arousal determines the intensity of the emotion, but it does not determine which emotion one experiences.

fear, anger, and disgust. He regarded emotions as endlessly varying, with no firm border between one and the next. Still, as far as it makes sense to label emotions, the James-Lange theory suggests that we distinguish our emotions by observing our bodily responses, especially our autonomic responses. We sense one pattern of autonomic responses when we are angry, another pattern when happy, still another when we are frightened or sad. (Remember, according to this theory, an emotion is the *perception* of what is happening in the body, not the *cause* of that change.) But is the autonomic state associated with anger noticeably different from the state associated with anxiety or any other emotion?

Heart rate and respiration increase during the experience of almost any emotion. Beyond that basic similarity, each emotional state has certain distinguishing physiological features (Levenson, 1992). With its "butterflies in the stomach" sensation, anxiety is probably the most distinctive emotional experience (Neiss, 1988). Your heart rate increases a little more when you are angry or frightened than when you are happy; the temperature of your hands increases more when you are angry than when you feel any other emotion. Your facial muscles respond in different ways when you experience various emotions, even though you may not actually be smiling or frowning (Tassinari & Cacioppo, 1992).

Granted that emotional states produce different physiological states, can those differences account for the

differences in emotions, as James and Lange suggested? If you begin to breathe rapidly and your heart begins to race, do you decide whether you are angry or frightened by checking the temperature of your hands—or any other physiological indicator? Probably not. The physiological differences among emotions are small, probably too small to identify an emotional state accurately (Lang, 1994). In fact, most people cannot accurately report even the intensity of their autonomic arousal.

A somewhat more promising possibility is that we judge our emotions by monitoring our facial expressions, or that having a certain facial expression at least increases the probability of feeling the associated emotion. James Laird (1974) molded people's faces into a smile or a frown by telling them to contract first this muscle, then that one, without ever using the words *smile* or *frown*. He found that an induced smile made people more likely to feel happy and that an induced frown

TRY IT YOURSELF

made them more likely to feel sad or angry. You might try this yourself: Smile for a while; then frown for a while. Do you feel happier when you are smiling, even though it was a purely voluntary smile?

If you do, remember the problem of *demand characteristics*: Maybe you say you felt happier when you smiled, just because you knew you were supposed to. Similarly, participants in an experiment often report what they think the experimenter expects them to report. Even though Laird never used the words *smile* or *frown*, the subjects may have identified their expressions and guessed that they were supposed to be related to their mood.

In another study, the experimenters found a clever way to conceal their purpose. They told subjects that the experiment had to do with how people with disabilities learn to write after losing control of their arms. The subjects were told to hold a pen either with their teeth or with their protruded lips, as Figure 12.10 shows. Then they were to use the pen in various ways, such as drawing lines between dots and making

checkmarks to rate the funniness of cartoons. When they held the pen with their teeth, their face was forced into a near-smile and they rated the cartoons as very funny. When they held the pen with protruded lips, they rated the cartoons as significantly less funny (Strack, Martin, & Stepper, 1988). (Try holding a pen one way and then the other, while reading newspaper cartoons. Do you notice any difference?)

The Cannon-Bard Theory of Emotions

Walter Cannon (1927), a prominent American physiologist, argued against the idea that autonomic responses were essential to emotion. Instead, he said, emotions produce autonomic changes and the cognitive experience of emotion independently. This view, as modified by Philip Bard (1934), is known as the **Cannon-Bard theory of emotions** (Figure 12.9). According to this theory, certain areas of the brain evaluate sensory information and, when appropriate, send one set of impulses to the autonomic nervous system and another set to the forebrain, which is responsible for the subjective and cognitive aspects of emotion.

The key assumption here is that the cognitive aspect of emotions is independent of the autonomic aspect. This assumption is only partly true. After spinal cord damage, people experience little sensation from their heart beating, stomach fluttering, and other autonomic responses; yet, they continue to experience fear, anger, and other emotions. They may report, however, that their emotions feel less intense than they did before. For example, one man with spinal cord damage reported that he occasionally *acted* angry without actually feeling angry, just because he knew that, if he did not respond to the situation with anger, other people would take advantage of him (Hohmann, 1966).

Similarly, after people with an intact nervous system take tranquilizing drugs, they generally report that their emotions, especially anxiety, seem less intense than before.



a



b

FIGURE 12.10 Facial expression can influence mood. When people hold a pen with their teeth (a), they rate cartoons as funnier than when they hold a pen with their protruded lips (b).

That is, the cognitive aspect of emotions may be partly independent of the autonomic changes, but it is not as independent as Cannon and Bard originally suggested.

Schachter and Singer's Theory of Emotions

Suppose we wire you to another person so that you share the other person's heart rate, breathing rate, skin temperature, and muscle tension. When the other person feels a particular emotion, will you feel it, too?

We cannot perform that experiment, but we can do the next best thing: We can use a drug to induce nearly the same physiological state in two people, and then see whether they both report the same emotion. To make things a little more interesting, we can put them in different situations. If emotion depends only on a person's physiological state, then both people will report the same emotion, even if they happen to be in different situations. Stanley Schachter and Jerome Singer (1962) put these ideas to the test in a now-famous experiment.

Schachter and Singer gave injections of the hormone epinephrine (adrenaline) to a group of college students who agreed to participate in the experiment. (Epinephrine mimics the effects of arousal of the sympathetic nervous system for about 20 to 30 minutes.) The experimenters told some of the subjects that the injections were vitamins; they did not warn them about the likely autonomic effects. Others were told to expect increased heart rate, butterflies in the stomach, and so forth. (Therefore, when these people did have such experiences, they would attribute the effects to the injection, not to the situation.)

Subjects were then placed in different situations. Some were placed, one at a time, in a situation designed to arouse euphoria (excited happiness). The others were placed in a situation designed to arouse anger.

Each student in the euphoria situation was asked to wait in a room with a very playful confederate, or accomplice, of the experimenter. The confederate flipped wads of paper into a trash can, sailed paper airplanes, built a tower with manila folders, shot paper wads at the tower with a rubber band, and played with a hula hoop. He encouraged the subject to join him in play.

Each subject in the anger situation was put in a waiting room with an angry confederate and asked to fill out an insulting questionnaire that included such items as these:

Which member of your immediate family does not bathe or wash regularly?

With how many men (other than your father) has your mother had extramarital relationships?

4 or fewer 5–9 10 or more

Most students in the euphoria situation showed strong emotional responses. Some joined the confederate in his play (Figure 12.11), and some of them initiated play of their own. (One jumped up and down on the desk, and another

opened a window and threw paper wads at passersby.) The students in the anger situation responded in various ways; some muttered angry comments, and a few refused to complete the questionnaire.

But another factor was important in this experiment. Some of the subjects had been informed beforehand that the injections would produce certain autonomic effects, including hand tremors and increased heart rate. No matter which situation those subjects were in, they showed only slight emotional responses. When they felt themselves sweating and their hands trembling, they said to themselves, "Aha! I'm getting the side effects, just as they said I would."

What can we conclude from this experiment? According to **Schachter and Singer's theory of emotions**, a physiological state is not the same thing as an emotion (see Figure 12.9). The intensity of the physiological state—that is, the degree of sympathetic nervous system arousal—determines the intensity of the emotion, but not the type of emotion. Depending on all the information people have about themselves and the situation, they can interpret a particular type of arousal as either anger, euphoria, or just an interesting side effect from taking a pill. That is, arousal intensifies an emotion, but cognitive appraisal of the situation tells us *which* emotion we are feeling. Table 12.1 contrasts Schachter and Singer's theory with the James-Lange and Cannon-Bard theories.

Unfortunately, Schachter and Singer's conclusions neglect another group of subjects—subjects whose results raise problems for their theory. These subjects, who were given placebo injections instead of epinephrine, showed about as much euphoria in the euphoria situation and as much anger in the anger situation as did the subjects injected with epinephrine. Therefore, critics argue, the epinephrine injections may have had nothing to do with the results. If we accept that possibility, we are left with this summary of the results from Schachter and Singer's experiment: People in a situation designed to induce euphoria act happy; people in an anger situation act angry. This result is not very interesting (Plutchik & Ax, 1967).

Schachter and Singer were right to call attention to the importance of cognition, but they may have ventured too far. By emphasizing how cognition determines our emotions, they downplayed the contributions of physiological states. To some extent, fear, anger, and happiness really do feel different physiologically, and those physiological differences contribute to the experience of our emotions.

The overall questions related to the James-Lange, Cannon-Bard, and Schachter-Singer theories are complex and difficult to investigate. Most investigators today believe that a more fruitful strategy is to explore the nature of specific emotions and their causes and expressions. Perhaps some day, after we know enough about specific emotions, we can return to study the issue of emotions in general.

CONCEPT CHECK

3. You are in a small boat far from shore, and you see a storm approaching. You feel frightened and start to

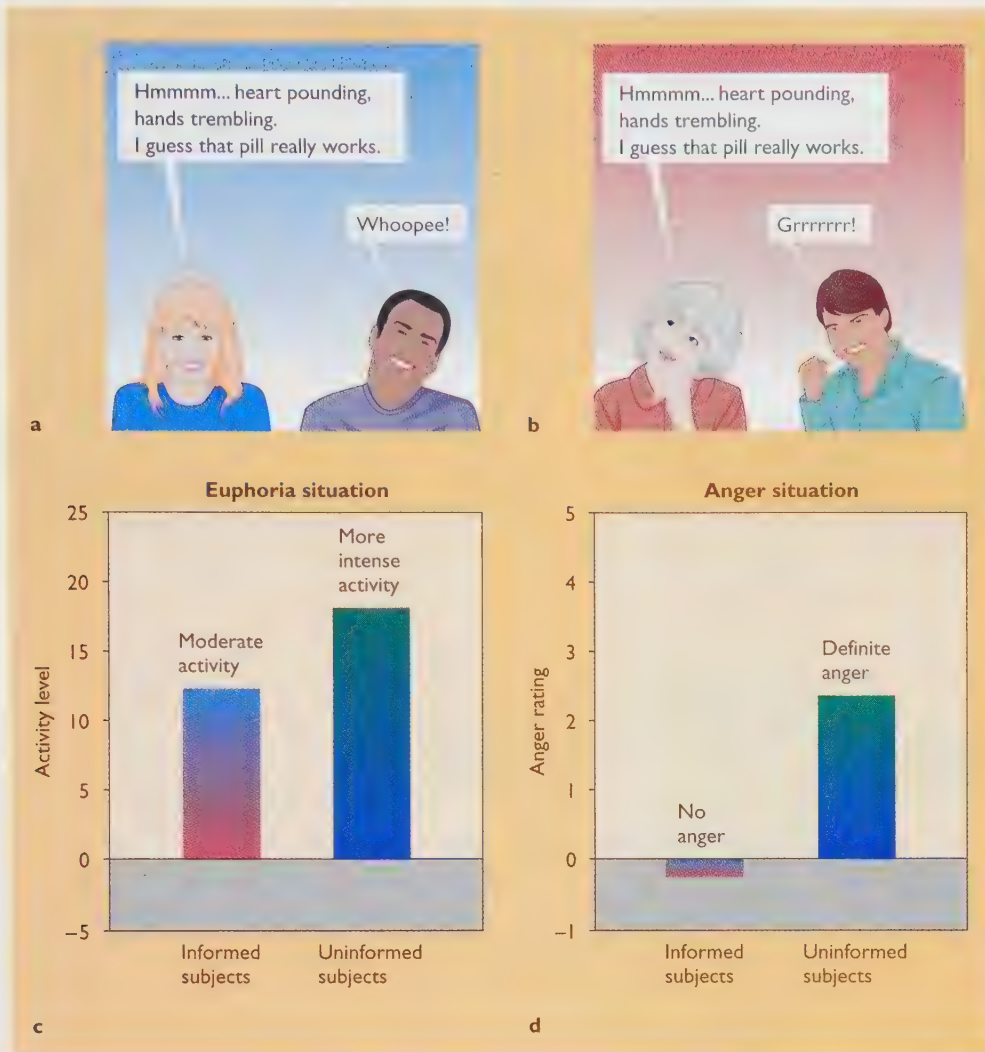


FIGURE 12.11 In Schachter and Singer's experiment, people who were uninformed about the effects of epinephrine reported strong emotions appropriate to their situation. According to Schachter and Singer, the autonomic arousal controls the strength of the emotion, but cognitive factors tell us which emotion we are experiencing.

tremble. According to the James-Lange theory, which came first—the fright or the trembling? According to Schachter and Singer's theory, which came first? (Check your answers on page 452.)

The Range of Emotions

How many different emotions do humans experience? Do we perhaps have a few “basic” emotions that combine to form other emotional experiences, just as three primary colors combine to produce all the other colors we see?

Psychologists have not yet agreed on a single list of basic emotions. Some psychologists have proposed a very short list, such as “pain and pleasure” or “happiness, sadness, and anger.” Most theorists who have tried to list the basic emotions have included happiness, sadness, anger, fear, disgust, surprise, contempt, and shame. However, most also include one or more other emotions, such as guilt, interest, hope, pride, relief, frustration, love, awe, boredom, jealousy, regret, or embarrassment (Keltner & Buswell, 1997). Still other theorists have questioned the whole con-

cept of basic emotions (e.g., Ortony & Turner, 1990). For example, anger, which is ordinarily considered a basic emotion, may be a combination of more primitive elements—widening the eyelids, lowering the eyebrows, pursing the lips, and making a fist.

How can we decide what is a basic emotion? The English language draws a distinction among shame, guilt, and embarrassment, but using different words does not guarantee that each is a distinct and basic emotion. English also distinguishes among red, orange, and pink (as well as scarlet, crimson, and other variants), but we do not accept each of these as a basic color. Psychologists have generally accepted the following criteria for establishing a basic emotion:

- Basic emotions should emerge early in life, without requiring much experience. For example, *fear* and *anger* are reasonable candidates to be considered basic emotions, because we see them even in infants. *Nostalgia* and *pride*, which emerge later in life, are weaker candidates to be considered basic emotions (Lewis, 1995).
- The basic emotions should be similar for people in different cultures. For example, if happiness and sadness are basic emotions, then people from one culture should easily

TABLE 12.1 Three Theories of Emotion

THEORY		COMMENT
<i>James-Lange theory</i>	<pre> graph LR AS[Anger situation] --> DPS1[Distinctive physiological state] DPS1 --> A[Anger] ES[Euphoria situation] --> DPS2[Distinctive physiological state] DPS2 --> E[Euphoria] </pre>	The situation determines the physiological state, and the physiological state completely determines the emotion.
<i>Cannon-Bard theory</i>	<pre> graph LR AS[Anger situation] --> CA[Anger cognitive appraisal] AS --> PS[Anger physiological state] CA --> A[Anger] PS --> A ES[Euphoria situation] --> CE[Euphoria cognitive appraisal] ES --> PE[Euphoria physiological state] CE --> E[Euphoria] PE --> E </pre>	The situation determines the cognitive state and the physiological state, independently.
<i>Schachter-Singer theory</i>	<pre> graph LR AS[Anger situation] --> CA[Anger cognitive appraisal] AS --> PA[Undistinctive physiological arousal] CA --> A[Anger] PA --> IE1[Intensity of whatever emotion] ES[Euphoria situation] --> CE[Euphoria cognitive appraisal] ES --> PA2[Undistinctive physiological arousal] CE --> E[Euphoria] PA2 --> IE2[Intensity of whatever emotion] </pre>	The situation determines the cognitive appraisal, which determines the emotion. All emotions are presumed to be physiologically indistinguishable; the physiological arousal determines the intensity of emotion but not the type of emotion.

recognize expressions of happiness and sadness by people of another culture. Cultures could differ, however, in their rules for such matters as when people should laugh and cry, what events should provoke jealousy, how much fear is normal, and how often people should get angry at others instead of taking the blame upon themselves (Mesquita & Frijda, 1992; Roseman, Dhawan, Rettek, Naidu, & Thapa, 1995).

- Basic emotions should have distinct biological bases. That is, a basic emotion probably depends on its own system of brain activity and triggers its own pattern of body activity, including perhaps a facial expression.

Producing Facial Expressions

Does each emotion have its own special expression? And why do we have facial expressions of emotions, anyway?

Quite simply, the function of facial expressions is communication. All primates (humans, apes, and monkeys) communicate their emotional states through gestures and facial expressions (Redican, 1982). (See Figure 12.12.) We humans can use spoken language as well, but we ordinarily prefer to use nonverbal expressions. You wink, nod, or smile to show a possible romantic interest; you withhold



FIGURE 12.12 The facial expressions of chimpanzees are similar to those of humans.

such expressions to indicate a lack of interest. (You *could* actually tell a stranger “I find you sexually attractive” or “please don’t approach me.” If you try this direct verbal approach, you will probably discover why most people prefer to rely on facial expressions.)

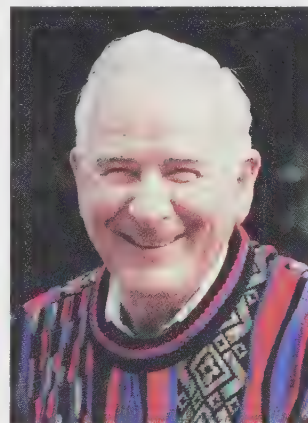
Facial expressions provide a kind of “truth in advertising.” That is, your expression reflects your internal state—such as being happy, sad, or angry—and therefore tells other people whether you are likely to interact with them in a friendly or an uncooperative manner (Buck, 1994). Emotional expressions occur mostly in the presence of other people. For example, one study found that people bowling with friends often smiled after making a strike or a spare; people bowling alone seldom smiled (Kraut & Johnston, 1979). In another study, Olympic medal winners smiled often while waiting for the awards ceremony with others but seldom when they were waiting alone (Fernández-Dols & Ruiz-Belda, 1997). Even 10-month-old infants smile more when their mothers are watching than when their mothers are sitting nearby but reading a magazine (Jones, Collins, & Hong, 1991). Laughter occurs vastly more often in social settings than when people are alone, and it is highly contagious (Provine, 1996, 1997). That is, we laugh mostly when others are laughing. Many televised comedies include a laugh track to help the viewers at home start laughing.

Voluntary smiles, frowns, and other facial expressions do not exactly match the full, spontaneous expressions (except in the case of skilled actors). For example, the smile of a truly happy person includes movements of the mouth muscles and the muscles surrounding the eyes (Figure 12.13a). Voluntary smiles (Figure 12.13b) include the mouth movements but generally do not include the muscles around the eyes (Ekman & Davidson, 1993). The full expression including the muscles around the eyes is called the **Duchenne smile**, named after Duchenne de Boulogne, the first person to describe it.

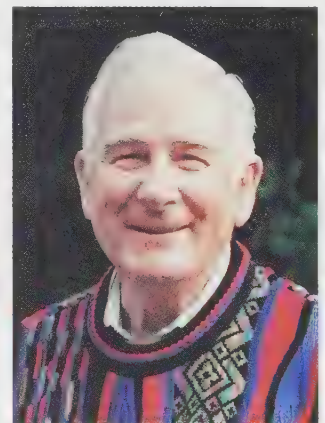
Understanding Facial Expressions

Do we learn how to make appropriate facial expressions, or are they part of our biological heritage? One way to approach this question is through naturalistic observations. Charles Darwin (1872/1965) asked missionaries and other people stationed in remote parts of the world to describe the facial expressions of the people who lived there. He found that people everywhere had similar facial expressions, including expressions of grief, determination, anger, surprise, terror, and disgust.

One century later, Irenäus Eibl-Eibesfeldt (1973, 1974) photographed people in different cultures, documenting the similarities in their facial expressions. He found smiling, frowning, laughing, and crying present



a



b

FIGURE 12.13 A spontaneous, happy smile (a) includes movements of both the mouth muscles and the muscles surrounding the eyes. This expression is sometimes called the “Duchenne smile.” A voluntary smile (b) ordinarily includes only the mouth muscles. Most people cannot voluntarily control the eye muscles associated with the Duchenne smile.



FIGURE 12.14 This laughing girl was born deaf and blind. (From Eibl-Eibesfeldt, 1973.)



FIGURE 12.15 A boy, blind since birth, covers his face in embarrassment. (From Eibl-Eibesfeldt, 1973.) He prevents others from seeing his face, even though he has never experienced sight himself.

throughout the world, even in children who are born deaf and blind (Figures 12.14 and 12.15). Evidently, at least some of our facial expressions develop spontaneously, without any need for imitation. Eibl-Eibesfeldt also found that people everywhere expressed a friendly greeting by briefly raising their eyebrows (Figure 12.16). The mean duration of

that expression is the same in all cultures: one third of a second from start to finish, including one sixth of a second in the fully elevated position.

To move beyond naturalistic observations, researchers typically use photos of people showing the presumed “basic” facial expressions, as shown in Figure 12.17. Try identifying these expressions yourself. First, look at each face and try to name its expression. (Please try now.)

Then look at each and match the expressions to the following labels (one face per label): anger, disgust, fear, happiness, sadness, and surprise.

Most people can match all six of the labels accurately. After researchers translate the labels into other languages, people in other cultures can also match them quite accurately, though not perfectly (Ekman, 1992). In other experiments, this procedure is reversed, and Europeans or Americans are asked to identify the facial expressions of people from other parts of the world. Here, again, the accuracy is fairly good.

What can we conclude from such data? We can conclude that people’s facial expressions have approximately the same meaning throughout the world (Ekman, 1994). However, the results do not necessarily indicate that people have precisely six basic emotions (Russell, 1994). One problem is that most expressions are somewhat ambiguous. For example, if you had looked only at face f, would you have been sure that it expressed fear? Using a matching procedure, you can label it as fear because it looks more like fear than any of the others do, and you can find other expressions to fit the other labels. But if you looked at f by itself, you might have called the expression sadness, anger, or even surprise. If you looked at d by itself, you might have called the expression fear or hope instead of surprise. In real

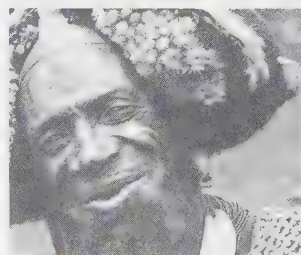


FIGURE 12.16 People throughout the world, such as this man from New Guinea, raise their eyebrows as a greeting. Traveling around the world, you could communicate anywhere by using such universal facial expressions of emotion as smiling and frowning. But you might give the wrong message if you didn’t know that nodding your head in some cultures means “no” or which hand and foot gestures are insulting. (From Eibl-Eibesfeldt, 1973.)



FIGURE 12.17 Paul Ekman has used these faces in experiments testing people's ability to recognize expressions of emotions. Can you identify which face conveys anger, disgust, fear, happiness, sadness, and surprise? Check the answer on page 452. (From Ekman & Friesen, 1984.)

life, we identify people's emotions only partly by their facial expression and also partly by the situation (Russell, 1997).

Furthermore, the fact that people of all cultures can recognize facial expressions for six emotions does not necessarily mean that these are basic emotions, much less that they are the *only* basic emotions. Consider an analogy to color: No doubt people throughout the world could match an array of colors to the names red, green, yellow, blue, pink, orange, brown, and purple, and probably even light green, dark green, yellowish green, and so forth. We do not conclude from this evidence that all those colors are "basic." Similarly, agreement on the label of a facial expression is weak evidence for calling the underlying emotion basic.

A Biological Approach to Identifying Basic Emotions

If people do have basic emotions, one way to identify them would be to find a brain area that is active during only one kind of emotion. For example, if researchers found some kind of brain damage that eliminated one kind of emotion without having much effect on the others, we could conclude that the eliminated emotion was a basic one, independent of the others.

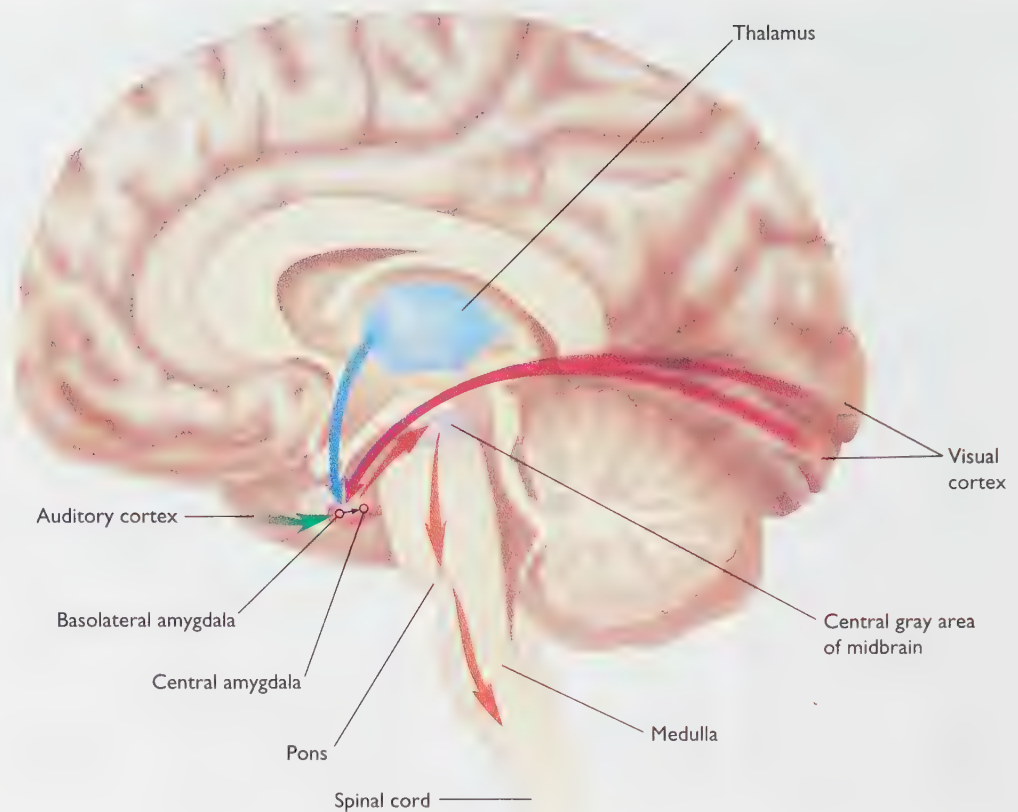
Researchers have found evidence linking fear or anxiety to part of the amygdala, as shown in Figure 12.18. This figure shows a human brain, although much of the research has been conducted with rodents. A sudden loud noise will evoke a startle response (tensing the muscles, etc.) in practically any mammal—human, rat, whatever—unless the individual

is deaf or unless damage has occurred to certain areas of the pons and medulla. Ordinarily, this startle response is even greater in anxiety-provoking situations. For example, people show a greater startle response in a dark room than in one that is well-lit. Just attaching shock electrodes to the skin enhances the startle response, even before any shocks are delivered (Grillon & Ameli, 1998). However, after damage to the amygdala in either humans or rats, many frightening stimuli no longer produce much of a fear response on their own and no longer enhance the startle response to a sudden noise (LaBar, LeDoux, Spencer, & Phelps, 1995; Phillips & LeDoux, 1992).

People who have suffered damage to the amygdala report feeling few fears or anxieties. They also cannot recognize other people's facial expressions of fear (Broks et al., 1998) and have trouble inferring other people's emotions from their tone of voice (Scott et al., 1997). However, their problems are not limited to depictions of fear or anxiety; they also have some trouble recognizing expressions of anger, disgust, and surprise.

Research has also identified a brain area apparently associated with disgust. In one study, people looked at photos of other people's facial expressions. Whenever they looked at photos showing disgust, activity increased in a brain area known as the anterior insular cortex (Phillips et al., 1997). That finding is particularly interesting because the anterior insular cortex is also known to respond strongly when people taste something offensive. Taste is also known as *gustation*, so *dis-gust* is literally *bad taste*. Our brains actually respond to disgusting things as if they tasted bad.

FIGURE 12.18 Certain small structures in the pons and medulla control unlearned fear reactions, such as a startle response to an unexpected loud sound. Damage to those areas can eliminate fear responses. Another structure, the amygdala, sends information to the pons and medulla. Damage to the amygdala eliminates learned fears but does not affect the automatic startle response to a loud sound. This diagram shows the human brain, although the relevant experiments were conducted with rats.



Happiness

What makes you happy? What makes you unhappy? The answers are more elusive than you might expect. The following quote is from Leo Tolstoy, the author of *War and Peace*, *Anna Karenina*, and other famous novels. At the time of this writing, Tolstoy was rich and famous but desperately unhappy:

I wandered about in the forest of human knowledge. . . . From one branch of human knowledge I received an endless number of precise answers to questions I had not asked, answers concerning the chemical composition of the stars, the movement of the sun toward the constellation Hercules, the origin of the species and of man, the forms of infinitely small atoms. . . . But the answer given by this branch of knowledge to my question about the meaning of my life was only this: You . . . are a temporary, random conglomeration of particles. The thing that you have been led to refer to as your life is simply the mutual interaction and alteration of these particles. This conglomeration will continue for a certain period of time; then the interaction of these particles will come to a halt, and the thing you call your life will come to an end and with it all your questions. You are a little lump of something randomly stuck together. The lump decomposes. . . . There is no more to be said.

My life came to a stop. . . . There was no life in me because I had no desires whose satisfaction I would have found reasonable. . . . If a fairy had come and offered to fulfill my every wish, I would not have known what to wish for. . . . I

did not even want to discover truth anymore because I had guessed what it was. The truth was that life is meaningless. (Tolstoy, 1882/1983, pp. 40–41, 27–28)

What would it take to make Tolstoy happy? Fame, wealth, family, and friends had failed to bring him happiness. If he were alive today, no doubt someone would recommend giving him antidepressant drugs. But I wouldn't count on drugs to do him much good either. What Tolstoy needed was a belief that the universe made sense and that life had meaning. Eventually, he found contentment, if not exactly happiness, by seeking spiritual (though not church-oriented) values.

TRY IT YOURSELF How happy would you rate yourself on a scale from 0 to 10, where 10 is extremely happy, 0 is extremely unhappy, and 5 is average? Then answer two more questions: What do you think would make you happier than you are now? And what *does* make you happy, on a typical day? (Please try answering these questions before you read on.)

In almost all countries except those that are war-torn or overrun with famine and disease, most people rate themselves somewhat above average in happiness (Diener & Diener, 1996). If you are like many people, your answer to “what would make you happier” probably included something like money, possessions, a good job, a boyfriend or girlfriend (or a “better” boyfriend or girlfriend), or more leisure time. However, the things that people think would make them happy generally do not keep them happy for

long. Lottery winners report great happiness shortly after winning but return to their usual happiness level (or even less!) within weeks or months. To the question “what does make you happy?” almost no one mentions money or possessions. Most people include spending time with family or friends, enjoying nature, doing something well enough to get a sense of accomplishment, or a system of beliefs.

In one study, more than 2,000 twins in Minnesota rated their own happiness. Their ratings correlated only slightly with their wealth, education, job prestige, marital status, or religious commitment, but in most cases, either both twins were happy, both were neutral, or both were unhappy (Lykken & Tellegen, 1996). We should nurture a little skepticism here: Maybe how happy people say they are isn't how happy they *actually* are; after all, how could we check the accuracy of someone's happiness report? Nevertheless, whatever it is that these happiness reports measure is apparently part of someone's personality, more rooted in genetics or the prenatal environment than it is in people's recent experiences. Some people tend to be gloomy and others tend to be cheerful, and most events shift the level of happiness only temporarily.

THE MESSAGE

Determining the Range of Emotions

When psychologists debate about how many basic emotions people have (if any), it may seem to you that the answer to this question is not terribly important. In a practical sense, it may not be, but it is part of a much larger issue. Similarly, in a practical sense, it may not matter whether there are 92 naturally occurring chemical elements or some other number, but when chemists established the number with confidence, they knew that they then understood something important about elements and that they were on the way to still greater understanding. Any field must start out by understanding its elements as clearly as possible. The psychology of emotions continues to be a difficult and sometimes frustrating field, simply because researchers are still searching for the answers to some basic questions.

S U M M A R Y

★ **The usefulness of emotions.** We make many decisions by imagining the emotional consequences of the possible outcomes. Brain-damaged people who suffer a loss of emotions also have trouble making and executing long-term plans. (page 435)

★ **Emotional intelligence.** In social situations, people need special skills to judge other people's emotions and intentions and the probable emotional outcomes of their own possible behaviors. People who misjudge others' emotions can easily cause themselves trouble. (page 436)

★ **Emotions and autonomic arousal.** Emotions are generally associated with arousal of either the sympathetic or parasympathetic branch of the autonomic nervous system. The sympathetic nervous system readies the body for emergency action. The parasympathetic nervous system promotes digestion and other less vigorous activities. (page 437)

★ **Opponent-process principle.** Removing the impetus for a given emotion instigates a sudden swing to the opposite emotion. For example, removing something that had made you happy would make you unhappy. (page 438)

★ **Polygraph.** The polygraph measures the activity of the sympathetic nervous system through such variables as heart rate, breathing rate, blood pressure, and electrical conductance of the skin. The polygraph is sometimes used as a “lie detector,” although it is not very accurate for that purpose. (page 439)

★ **James-Lange theory.** According to the James-Lange theory of emotions, an emotion is the perception of a change in the body's physiological state. (page 442)

★ **Cannon-Bard theory.** According to the Cannon-Bard theory, the cognitive experience of an emotion occurs at the same time as physiological arousal, but the two are independent of each other. (page 443)

★ **Schachter and Singer's theory.** According to Schachter and Singer's theory, autonomic arousal determines the intensity of an emotion but does not determine what that emotion will be. We identify an emotion on the basis of how we perceive the situation. (page 444)

★ **Range of emotions.** Psychologists do not fully agree about which emotions, if any, are basic. (page 445)

★ **Facial expressions.** Many human facial expressions have similar meanings in cultures throughout the world. (page 446)

★ **Measuring happiness.** Happiness is difficult to measure, and most of what we know about it is based on correlational data. Material possessions alone cannot guarantee happiness. Happiness level is a fairly stable aspect of personality despite changes in people's lives. (page 450)

Suggestions for Further Reading

- Damasio, A.R. (1994). *Descartes' error*. New York: G.P. Putnam's Sons. Argues that emotion is a necessary contributor to intelligent decision-making.
- LeDoux, J.E. (1996). *The emotional brain*. New York: Simon & Schuster. Discussion of the biological bases of emotion, by a pioneering researcher.
- Russell, J.A., & Fernández-Dols (Eds.) (1997). *The psychology of facial expression* (pp. 295–320). Cambridge, England: Cambridge University Press. Good review of research on how we express emotions.

Terms

emotional intelligence the ability to perceive, imagine, and understand emotions and to use that information in decision-making (page 436)

autonomic nervous system a section of the nervous system that controls the functioning of the internal organs (page 437)

sympathetic nervous system a system composed of two chains of neuron clusters lying just to the left and right of the spinal cord; these neurons send messages to the internal organs to prepare them for a burst of vigorous activity (page 437)

parasympathetic nervous system a system of neurons located in the medulla and the bottom of the spinal cord; these neurons send messages to the internal organs to prepare the body for digestion and related processes (page 437)

opponent-process principle of emotions the principle that the removal of a stimulus that excites one emotion causes a swing to an opposite emotion (page 438)

polygraph a machine that simultaneously measures heart rate, breathing rate, blood pressure, and electrical conduction of the skin (page 439)

guilty-knowledge test a test that uses the polygraph to measure whether a person has information that should be known only by someone guilty of a certain crime or someone who talked with the guilty person (page 441)

James-Lange theory the theory that emotion is merely our perception of autonomic changes and movements evoked directly by various stimuli (page 442)

Cannon-Bard theory of emotions the theory that certain areas of the brain evaluate sensory information and, when appropriate, send one set of impulses to the autonomic nervous system and another set to the forebrain, which is responsible for the subjective and cognitive aspects of emotion (page 443)

Schachter and Singer's theory of emotions the theory that the intensity of sympathetic arousal determines the intensity of an emotion but that cognitive factors determine the type of emotion. (page 444)

Duchenne smile a spontaneous expression that includes movement of both the mouth muscles and certain muscles near the eyes (page 447)

2. After someone takes a drug repeatedly, the A state becomes weaker. (That is known as tolerance, as discussed in Chapter 7.) The B state (withdrawal) becomes stronger. (page 439)
3. According to the James-Lange theory, the trembling and shaking came first. Schachter and Singer's theory agrees. However, according to the James-Lange theory, your perception of the trembling and shaking leads immediately and automatically to the experience of fear. According to Schachter and Singer's theory, you first interpret your trembling on the basis of the circumstances before you experience fear: "Am I shaking because of that pill I took? Because someone made me angry? Because I'm excited? Because I'm frightened?" (page 444)

Answers to Other Question in the Text

Figure 12.17, page 449: From left to right, top to bottom, the faces express happiness, anger, sadness, surprise, disgust, and fear.

Web Resources

What's Your EQ?

www.utne.com/cgi-bin/eq/

How well do you know or recognize your feelings and use them to make effective life decisions?*

Drugs, Brains & Behavior

daniel.drew.edu/~ctimmons/drugs/chap04.htm#James-Lange
Theory of Emotion

C. Robin Timmons of Drew University and Leonard W. Hamilton of Rutgers University briefly describe the James-Lange, Cannon-Bard, and Schachter and Singer theories of emotions.

*Note: Any psychological test that is readily available to the public, while entertaining and possibly informative, should not be used to make important decisions about yourself or others. The most powerful, valid, and reliable psychological assessment devices are usually kept under the tight control of their creators or copyright holders.

Answers to Concept Checks

1. When you ride a roller coaster, your heart rate increases (sympathetic activity). After you get off the ride, your heart rate falls to lower than usual (rebound increase in parasympathetic activity). (page 439)

Anger and Violent Behaviors

What are the circumstances that lead to violence?

Why are some people more violent than others?

During World War II—while nearly all the industrialized nations were at war, while the Nazis were exterminating the Jews, and while the United States was preparing a nuclear bomb that it later dropped on Japan—Mohandas K. Gandhi, the world's foremost advocate of nonviolence, was in jail for leading a protest march against British rule in India. The charge against Gandhi was, ironically, "disturbing the peace." Someone asked Gandhi what he thought of Western civilization. He replied that he thought it might be a good idea.

Human beings are capable of ghastly acts of cruelty against one another. They are also capable of acts of nobility, heroism, and courageous opposition to violence, as in Gandhi's case. The struggle to understand violence is among the most important goals facing humanity in general, and psychology in particular.

Provocations of Anger and Violence

Most people become angry far more often than they actually attack anyone. Participants in one study kept an "anger diary" for a week (Averill, 1983). Most of their experiences with anger were provoked by people they knew well, such as, "My roommate locked me out of the room when I went to the shower. That's the third time. If he does it one more time, I'm gonna kill him." The diaries also included descriptions of what people did in response to each event. Nearly all the replies were either, "I talked to the person who made me angry," "I talked to someone else about it," or "I did nothing about it." Very few admitted to issuing even threats of violence, and certainly no one said, "The next day he locked me out again, so I killed him."

If you try keeping track of your own anger experiences, you will probably find that you are often provoked when

someone prevents you from doing something you wanted to do, such as if someone borrows something you needed without asking you. You will also probably find that few or none of your angry episodes lead to violence. Psychologists have long spoken of the **frustration-aggression hypothesis**, the idea that frustration—a failure to obtain something that one expected—leads to aggressive behavior (Dollard, Miller, Doob, Mowrer, & Sears, 1939). However, this hypothesis has several limitations. First, frustration produces anger and the potential for aggression mainly when one attributes an intention to the person who caused the frustration. For example, if someone runs down the hall, bumps into you, and knocks you down, you might feel angry. But if someone slipped on a wet spot on the floor and knocked you down, you would probably not feel angry, because you do not assess blame. Secondly, even when frustration leads to anger, the anger will not necessarily lead to aggression. Finally, frustration is important only for emotional aggression, not for calm aggressive behaviors that people learn as strategies for getting what they want.

Leonard Berkowitz (1983, 1989) therefore proposed a more comprehensive theory: Any unpleasant event—frustration, pain, foul odors, a hot environment, frightening information, whatever—excites both the impulse to fight and the impulse to flee. Which impulse dominates depends on the circumstances, and one of the main circumstances is the expected result of an attack. If you have frequently struck people before, have won most of your fights, and have seldom been punished, and you think the person who just spilled coffee on you looks easy to beat, you may threaten this person or even strike. But if you have had mostly bad experiences when you got into fights, or if the offending person looks large and ferocious, you will suppress your anger. And if the one who spilled the coffee is your boss, or the loan and scholarship officer at your college, you will most likely smile and apologize for being in the way.

Although any kind of frustration or disappointment can lead to anger, and potentially to violence, the likelihood is particularly strong in a sexual context. Throughout the animal kingdom, who are more likely to fight with one another—males or females? Males. When do they fight the most? During the reproductive season. And what are they fighting about? Access to females. In some cases, they are fighting for a position of dominance, but the main value of dominance is increased access to sexual partners. In humans as well, one of the leading causes of murder is sexual jealousy (Daly, Wilson, & Weghorst, 1982).

Characteristics of Relatively Violent People

Can we identify the people who are most likely to commit violent acts? In at least one regard, many psychologists were on the wrong track for a long time, assuming that people committed violence because of low self-esteem. According



Throughout the animal kingdom, aggressive behavior is common among males when they are competing for the attention and affection of females.



to that idea, people who thought little of themselves tried to build themselves up by tearing someone else down. Perhaps (but only perhaps) that hypothesis works in certain cases of group violence. A fair number of outbreaks of ethnic violence have occurred when a group has suffered defeats and humiliations and then embraced an ideology that blamed another group for their problems and promised them a glorious future after the destruction of the offending group (Staub, 1996).

For individual violence, however, the research supports little or no link between low self-esteem and violent behavior. The victims certainly report low self-esteem, sometimes extremely low, but the aggressors tend to be arrogant, egotistical, and quite self-confident. At most, it may be that violence results when something threatens someone's high self-esteem, and that person then strikes out against another person to reaffirm a sense of superiority (Baumeister, Smart, & Boden, 1996).

One extensive study of Norwegian and Swedish bullies, ages 7 through 16, found that hardly any of them suffered from feelings of anxiety or insecurity. They were as a rule large, strong, impulsive, and indifferent to their victims. Why did they do it? They said it was fun. They enjoyed taking another child's money, forcing him to eat weeds, leading him around as a pet with a string around his neck . . . and, most importantly, they got away with it. Psychologists found they could reduce the bullying by increasing adult supervision, enforcing rules, and having serious talks with the offending children and their parents (Olweus, 1995).

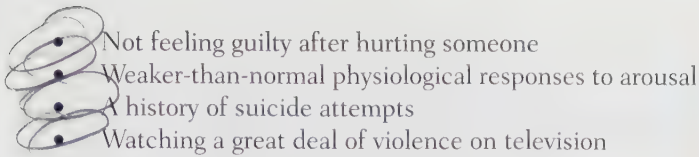
If low self-esteem is not a good predictor of violence, what is? Sometimes psychologists or other mental-health professionals are asked to predict how violent someone will be. For example, a judge may be trying to decide between a prison sentence and a period of probation for someone recently convicted of a crime. Or perhaps someone has been in prison and is now eligible for parole, or has been in a hospital for mentally disordered criminal offenders and is now being considered for release.

Under such circumstances, how accurate would you guess that a psychologist's predictions are? Predictions based on interviews are a bit more accurate than they would be by flipping a coin, but they are not impressive (Monahan, 1984). One reason for these mediocre results is that violent behavior is sporadic and situation-dependent. Even the most violence-prone people can be pleasant and charming, especially when they are talking to a psychologist who has the power to decide whether they belong in prison. The best basis for predicting violence is past behavior; people with a long history of violent acts are likely to commit some more, whereas those having committed only a single violent act are much less dangerous. From an examination of someone's past, it is possible to make moderately useful predictions, although it is still best to admit uncertainty, just as weather forecasters usually qualify their predictions, such as, "the chance of rain is 80% for Tuesday and 60% for Wednesday." Many psychologists prefer to say something like, "This defendant has a less than 25% risk of violence if he gets a job and a higher than 50% chance if he does not" (Borum, 1996; Grisso & Tomkins, 1996; Monahan & Steadman, 1996).

Specifically, people who were physically abused as children and who witnessed violence between their parents are more likely than others to commit repeated acts of violence, including murder (Malinosky-Rummell & Hansen, 1993). The physical pain of being beaten may provoke future violence, and a violent parent provides a model for the child to imitate. (Recall the principles of social learning from Chapter 6.) Even this predictor is far from foolproof, however; only about one third of abused children become abusive parents (Widom, 1989).

Several other factors are associated with a tendency toward violent behavior (Lewis et al., 1985; Lynam, 1996; Os-
ofsky, 1995; Raine, Venables, & Williams, 1990):

- A history of violent or impulsive acts and of frequent associations with antisocial individuals
- Growing up in a violent neighborhood



The probability of attack also depends on biological factors. In all cultures, violent and criminal activities are more common in men than in women, and the probability of such activities increases sharply during puberty, when testosterone levels are increasing. Figure 12.19 shows the probability of arrest for robbery, burglary, and aggravated assault as a function of age in the United States. For each crime, the age of the maximum number of arrests was set at 100%, so the graph does not show that burglary is more than three times more common than either of the other two crimes. Note that, for each crime, the rate of occurrence rises sharply during the teen years, reaches a peak at 17 to 21 years, and then declines steadily.

Figure 12.20 shows the percentage of male juvenile delinquents born in 1945 or 1958 who had their first arrest at various ages. Note that the probability of a first arrest increases sharply between the ages 12 and 16 and then drops (Tracy, Wolfgang, & Figlio, 1990). The probability of an arrest does not drop at age 17, but the probability of a first arrest does; that is, a boy who did not get himself into trouble before age 17 will probably not start later.

With humans as well as with other species, testosterone increases the probability of violence (Bernhardt, 1997; Brooks & Reddon, 1996; Dabbs, Carr, Frady, & Riad, 1995). (Recall the link between violence and sexual jealousy.) A variety of drugs, notably alcohol and tranquilizers, increase the probability of violent behavior (Bushman, 1993; Bushman & Cooper, 1990). Tranquilizers presumably increase violent behavior by decreasing people's fear of the possible retaliation by the person being attacked. Also, the probability of violent behavior is highest

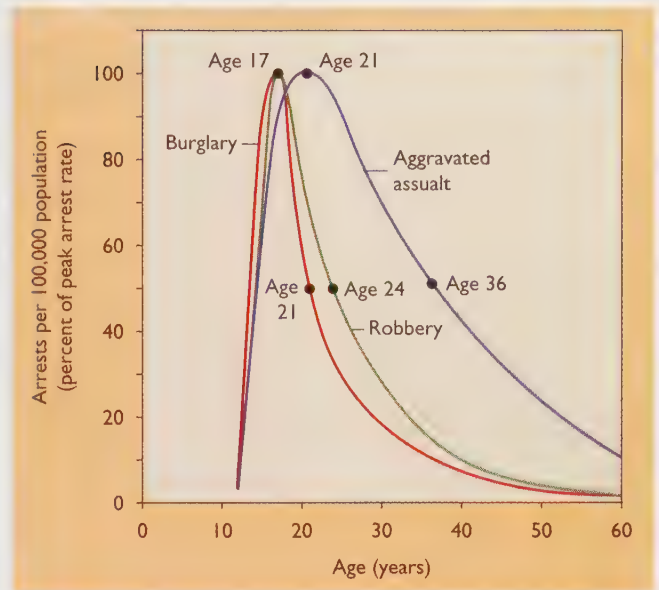


FIGURE 12.19 Relative frequencies of arrest at different ages in the United States. For each crime, the age of the maximum number of arrests was set at 100%. Note that the arrest rate rises for each crime until ages 17–21 and thereafter declines. (From data of the Federal Bureau of Investigation, 1984.)

among adults with low activity of the neurotransmitter serotonin. Exactly how a deficit of serotonin activity could lead to violence is unknown, but measurements of serotonin metabolites in the blood or urine have provided moderately accurate predictors of which paroled prisoners will be arrested again (Virkkunen, Eggert, Rawlings, & Linnoila, 1996) and which survivors of violent suicide attempts will attempt it again (Roy, DeJong, & Linnoila, 1989). Drugs that increase serotonin activity help to decrease aggressive acts among people with a history of such (Coccaro & Kavoussi, 1997).

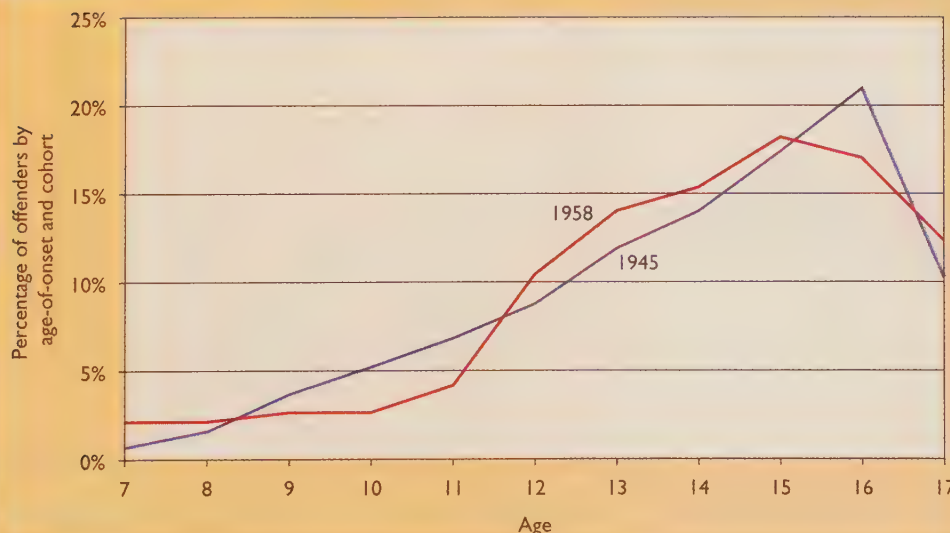
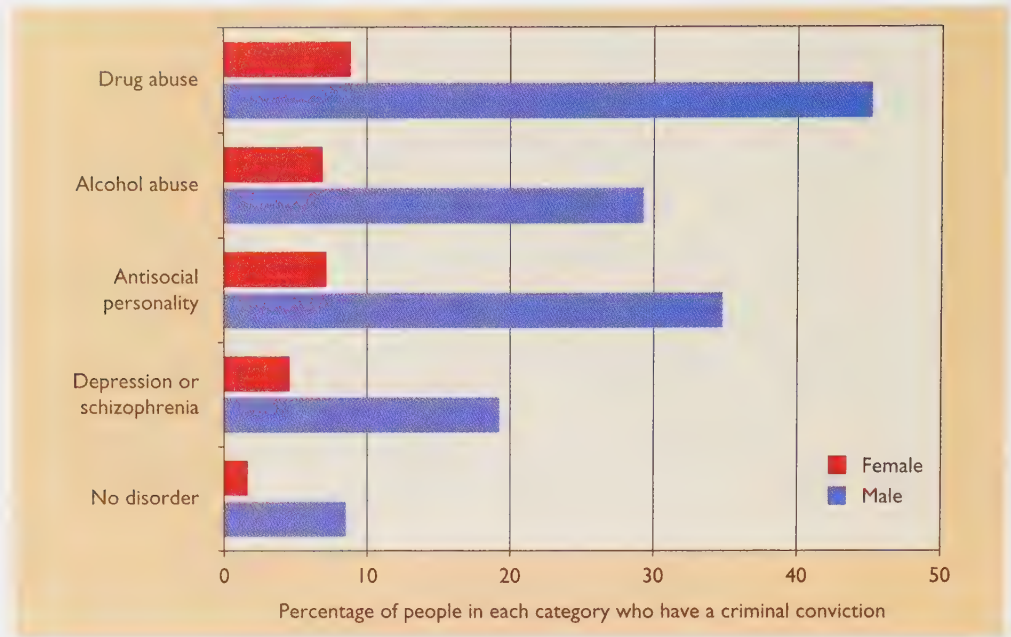


FIGURE 12.20 The percentage of juvenile delinquents in Philadelphia, Pennsylvania, who had their first arrest at each age. The number of total arrests does not decline at age 17, but the number of first-time arrests does. (From Tracy, Wolfgang, & Figlio, 1990.)

FIGURE 12.21 The percentage of people with various conditions who have at least one criminal conviction. (Based on results from hundreds of thousands of people in Denmark, from Hodgins, Mednick, Brennan, Schulsinger, & Engberg, 1996.)



Violence by Mental Patients

Decades ago, psychologists believed that mental patients or former mental patients were no more dangerous than anyone else and no more likely to commit crimes. However, that conclusion was based on distorted evidence. At the time, most mental patients who were admitted to mental hospitals stayed there indefinitely, so only the best recovered and least dangerous ones were released. Today, nearly all mental patients are released after brief stays, so our data are more representative.

The results vary from one study to another, but in general it appears that patients with most kinds of mental disorders are somewhat more likely than other people to be arrested for serious crimes (Eronen, Hakola, & Tiihonen, 1996; Hodgins, Mednick, Brennan, Schulsinger, & Engberg, 1996; Teplin, Abram, & McClelland, 1996). (See Figure 12.21.) A few (very few) deeply disturbed mental patients commit extreme, senseless acts of violence, such as pushing a stranger in front of a subway train in a large city, and thereby receive a disproportionate amount of publicity (Martell & Dietz, 1992).

One type of mental disorder that is associated with a high probability of crime is antisocial personality disorder (Hodgins et al., 1996). However, the criteria for diagnosing that condition include lack of remorse and repeatedly performing acts that are grounds for arrest (American Psychiatric Association, 1994), so its association with crime is hardly a mystery. The other mental disorder associated with a high crime rate is drug and alcohol abuse (Hodgins et al., 1996; Monahan, 1992; Teplin et al., 1996). One interesting, unresolved question is to what extent the violence is due to the drugs and to what extent it is a product of the kind of person who abuses drugs. Curiously, one study found that giving methylphenidate (Ritalin), a stimulant

drug, to adolescents with conduct problems actually decreased their troublesome behaviors (Klein et al., 1997).

Still, even though certain disorders are associated with an increased risk of violence, the task remains to predict which individuals with a given disorder are dangerous. As with anyone else, interviews and personality measures provide only weak predictors of violence (Bonta, Law, & Hanson, 1998). The best predictor is the person's past history of violent or criminal behavior. As shown in Figure 12.22,

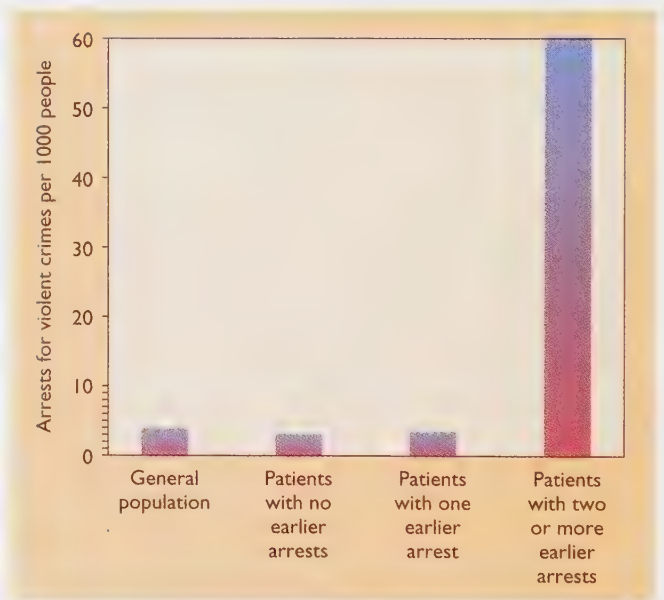


FIGURE 12.22 The arrest rates for mental patients and for the general public. Patients with a history of criminal offenses are likely to continue to be dangerous, but those without such a history are not. (After data of Cocozza, Melick, & Steadman, 1978.)

mental patients with a past history of criminal arrests are likely to be arrested again; those with no previous arrests are likely to continue keeping a clean record (Cocozza, Melick, & Steadman, 1978).

Sexual Violence

Rape is sexual contact obtained through violence or threats. Although that definition sounds simple enough, in practice, offenses range on a continuum from forcible rape through ambiguous resistance. For example, 9% of the women surveyed at 32 colleges reported that they had been forced into unwanted sexual intercourse, and 25% said they had participated in unwanted intercourse in response to verbal coaxing or while under the influence of alcohol (Koss, Gidycz, & Wisniewski, 1987; see Figure 12.23). At the same colleges, only 4.4% of the men said they had forced a woman to have sex (Koss & Dinero, 1988). Although it is possible that a small percentage of the men forced themselves sexually on a larger percentage of women, a more likely interpretation is that men and women interpreted the same events differently. In many cases, a couple had kissed and petted voluntarily; when the man wished to progress to intercourse, evidently the woman said no but the man either disbelieved or disregarded her refusal. Preventing date rape is partly a matter of persuading men to respect a woman's refusal and partly a matter of advising women to express their refusal clearly and firmly.

What kind of men commit rape? Unfortunately, most of our data are based on convicted rapists, who are generally those who committed violent attacks and who may therefore not be representative of other rapists. Still, the best available evidence indicates that most men who sexually attack women have a history of hostility and violence against both men and women (Malamuth, Sockloskie, Koss, & Tanaka, 1991). Many rapists were abused children; many feel anger toward women and a need to dominate or control them. Although much less is known about date rapists, one survey of admitted date rapists found that they also reported

anger toward women and a drive to dominate them (Lisak & Roth, 1988). The probability of rape is also elevated when a man becomes so intoxicated from alcohol or other drugs that he sheds the inhibitions that would ordinarily prevent such an attack (Lisak & Roth, 1988).

Just as psychologists do with other kinds of criminals, they sometimes try to predict whether convicted sex offenders (such as rapists and child molesters) are still dangerous—for example, when a sex offender is under consideration for parole. It would be pointless to ask sex offenders whether they are still attracted by the idea of rape or sexual contact with children; someone hoping for parole is not likely to answer honestly. The preferred approach is to present photos or audiotapes of various kinds and measure erections of the penis. Many rapists become sexually aroused by portrayals of rape, and many child molesters become aroused by photos of naked children, whereas most other men are not sexually aroused by either of these kinds of stimuli (Earls, 1988; Harris & Rice, 1996). However, although most psychologists feel comfortable in assuming that this procedure indicates whether someone is likely to commit rape or child abuse, we have little direct evidence to test this assumption.

What should a woman do if a man is trying to rape her? Some psychologists recommend fighting back; others recommend trying to appeal to the would-be rapist's better nature; others recommend doing something to upset or offend the rapist, such as vomiting. However, for each of these recommendations, one can find other psychologists who say it is a bad idea that is likely to backfire (Fischhoff, 1992). The problem is that no one approach is sure to succeed; the results depend on the woman, the man, and the setting.

SOMETHING TO THINK ABOUT

Rapists and child molesters sometimes pore over sexually explicit magazines and watch videotapes just before committing an offense (Malamuth & Donnerstein, 1982; Marshall, 1988). Can we conclude that such materials lead to the offenses? (Remember, correlation does not mean causation.) What kind of evidence would we need to determine whether sexually explicit materials lead to sexual offenses? ★

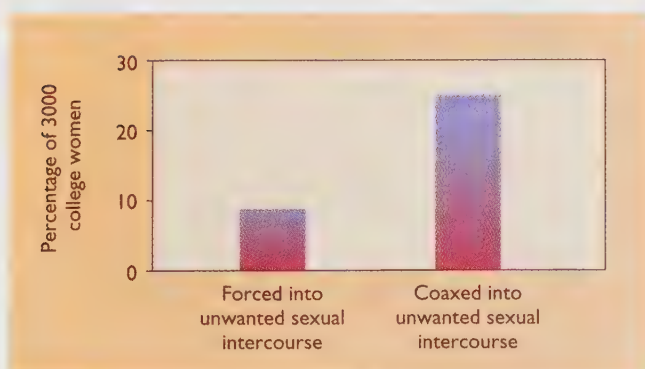


FIGURE 12.23 How common is date rape on college campuses? The answer depends on how we define rape.

THE MESSAGE Controlling Violence

Most of us would like to believe that people are fundamentally good and that acts of violence and cruelty are an abnormality: If only we could get rid of poverty, injustice, ignorance, low self-esteem, and so forth, then people would stop being cruel to one another. The problem with this noble belief is that some perpetrators of unspeakable cruelty have had all the advantages of wealth and opportunity. Improvements in society can decrease violent crime—after all, the crime

rate usually does go down when the economy improves—but we are unlikely to eliminate violence completely.

Is punishment therefore the answer? Only in part. Children are less likely to be bullies if they have enough adult supervision, and adults are also less likely to abuse others if they believe someone will retaliate against them. However, making punishments quicker and more certain is more effective than making them harsher. In fact, excessively painful punishment actually triggers aggressive behavior.

Finally, it is possible to help people control their aggressive tendencies, if they wish to receive help. People can be taught to accept frustration and to react to disappointments in less disruptive ways. Those who are accustomed to gaining attention or other reinforcements from violent outbursts can learn other ways to get what they want (Fehrenbach & Thelen, 1982). People are less likely to turn to violence, even as a “last resort,” if they can see other options.

S U M M A R Y

★ **Anger.** People experience anger frequently, although it seldom leads to violent acts. Aggressive behavior often occurs in defense of territory or in competition for a mate. Frustration or other unpleasant experiences often prompt aggressive behavior. (page ●●●)

★ **Predicting violence.** Psychologists and psychiatrists find it difficult to predict whether a particular prisoner would be dangerous, if released. The best way to make such predictions is to review someone's past behavior, especially the history of violent behavior. (page ●●●)

★ **Violence by mental patients.** People with certain kinds of psychological disorders, especially substance abusers, are more likely than other people to commit violent acts. Still, the best predictor of violence among mental patients, as among other people, is their past behavior. (page ●●●)

★ **Sexual violence.** Most of the men who are convicted of violent rape have a history of other kinds of violent acts also. (page ●●●)

Suggestion for Further Reading

Staub, E. (1989). *The roots of evil*. New York: Cambridge University Press. An inquiry into why people engage in extremely violent behavior.

Terms

frustration-aggression hypothesis the theory that frustration leads to aggressive behavior (page ●●●)

rape sexual contact obtained through violence or threats (page ●●●)

Web Resources

MINCAVA: Global Links on Violence

www.mincava.umn.edu/global.asp

The Minnesota Center Against Violence & Abuse (MINCAVA) maintains hundreds of links to sites that provide information about all forms of violence.

Health Psychology

How do our emotions affect our health?

Imagine you meet a man who is suffering from, say, multiple sclerosis. Would you say, "It's his own fault he's sick; he's being punished for his sins"? I presume you would neither say nor believe anything so cruel. However, in the Middle Ages and in ancient times, many people believed just that. We congratulate ourselves today on having advanced beyond that way of thinking; we know it is wrong to blame the victim.

Or do we? We may think that cigarette smokers are at least partly at fault if they develop lung cancer. We note that AIDS is usually contracted by people with a history of intravenous drug use or unsafe sex. If women drink alcohol during pregnancy, we hold them partly responsible for the deformities or mental retardation of their infants. As we learn more and more about the causes of various illnesses,



Health psychology deals with all the various ways that people's behavior affects their health, ranging from why some people neglect to take their prescribed medication to why some people persist in smoking, heavy drinking, and other risky behaviors.

we expect people to accept more responsibility for their own health, even if we do not exactly use the word *blame* when people become ill.

People's behavior does, in fact, influence their health. Unfortunately, we can easily overstate the extent of that influence. Some people are always as careful as possible about their diet, exercise regularly, and have healthy habits but develop serious illnesses anyway. Psychological factors influence our health, but these are not the only influences.

Health psychology is concerned with how people's behavior can enhance health and prevent illness and how behavior contributes to recovery from illness (Brannon & Feist, 1992). It deals with such issues as why people smoke, why they sometimes ignore their physician's advice, and how to reduce pain. In this section, we shall focus mainly on how stress and other emotional conditions affect health.

Stress

Have you ever gone without sleep several nights in a row trying to finish an assignment before a deadline? Or waited what seemed like forever for someone who was supposed to pick you up? Or had a close friend suddenly not want to see you anymore? Or tried to explain why you no longer want to date someone? Each of these experiences provokes an emotional response and causes stress.

Selye's Concept of Stress

According to Hans Selye (1979), an Austrian-born physician who worked at McGill University in Montreal, **stress is the nonspecific response of the body to any demand made upon it**. Every demand on the body evokes certain specific responses as well. The body responds in one way to the loss of blood, in another way to the lack of sleep. But all demands on the body evoke generalized responses such as increased response of the sympathetic nervous system, increased release of the hormone epinephrine, and impaired concentration.

When people say, "I've been under a lot of stress lately," they are generally referring to a string of unpleasant experiences. Selye's concept of stress is broader: He includes any experience that brings about change in a person's life. For example, getting married and being promoted are presumably pleasant experiences, but they also demand that you make changes in the way you live, so in Selye's sense, these events produce stress. **It is unclear, however, whether a favorable stressor makes the same demands on the body as does an unfavorable stressor. According to Selye, dealing with the effects of poverty, racism, sexism, or a lifelong disability would not count as stress, however; only changes in your life count as stress.**

Selye believed that the body goes through three stages in its response to a stressor: The first is **alarm, a brief period of high arousal of the sympathetic nervous system, readying**

the body for vigorous activity. Some stressors last longer than the body can maintain this high state of arousal, however. Perhaps you have a high-stress job or you live near the runway of a busy airport. If so, you cannot overcome your problem with a brief burst of intense activity. You enter **resistance, a stage of prolonged but moderate arousal.** Your epinephrine levels remain at a high level day after day, week after week (Figure 12.24). Your adrenal cortex secretes cortisol and several other hormones that elevate blood sugar and enhance metabolism. The increased fuel supply to the cells enables them to sustain a high, steady level of activity to endure prolonged stress. However, you no longer feel ready for vigorous activity; you feel withdrawn and inactive much of the time, your performance deteriorates, and you complain of a decreased quality of life (Evans, Bullinger, & Hygge, 1998).

If the stress is even more intense and long-lasting, the body enters the third stage, **exhaustion.** *As cortisol and other hormones shift energy toward increasing blood sugar and metabolism, they shift it away from synthesis of proteins, including the proteins necessary for activity of the immune system.* In the short term, that shift may not be a problem; however, severe stress over many months can weaken the immune system and leave an individual vulnerable to a variety of illnesses (Cohen, 1996). The end result is characterized by weakness, fatigue, loss of appetite, and a general lack of interest.

Posttraumatic Stress Disorder

Perhaps the most powerful demonstration of the effects of severe stress is **posttraumatic stress disorder (PTSD).** *People who have endured extreme stress feel prolonged anxiety and depression* (Pitman, Orr, Forgue, DeJong, & Claiborn, 1987). This condition has been recognized in postwar periods throughout history and given such names as “battle fatigue” or “shell shock.” One nationwide survey reported posttraumatic stress disorder in 20% of the American veterans who were wounded in Vietnam (Helzer, Robins, & McEnvoy, 1987). It also occurs in rape or assault victims, torture victims, survivors of an air-



This group of Vietnam vets met regularly for a year to work on problems related to posttraumatic stress. Since the group disbanded, one member has had an exhibition of his hand-colored photographs. (This group portrait is a sample of his work.) Another member has started his own company. Some members have created new careers; some have been in and out of substance-abuse programs. Most continue to experience vivid dreams full of war images.

plane crash or a severe automobile crash, and witnesses to a murder.

Anyone who suffers through a traumatic event experiences severe stress at the time and shortly afterward. People differ greatly, however, in their long-term reactions. For example, in 1980, a man kidnapped three college cheerleaders and raped and murdered one of them. The other two escaped and eventually testified against their assailant. One of the two quickly put the events behind her; she completed her education, married, got a job, had children, and only occasionally thought about her gruesome ordeal. The other woman suffered a clear case of PTSD, including long bouts of depression and intense flashbacks. She found herself unable to focus on her studies and became very uneasy in her relationships with men. At one point, she rented a room with a view of the prison, so that she could keep an eye on it, to make sure that her assailant didn't escape (Krueger & Neff, 1995). Why do some people develop PTSD, whereas others who had similar experiences do not? Psychologists do not yet know.

People with posttraumatic stress disorder can suffer from frequent nightmares, outbursts of anger, constant unhappiness, and guilt. The guilt is a special kind of experience, often called survivor's guilt, common to people who survive a catastrophe in which many other people died. People with PTSD may also have difficulty concentrating or relating emotionally to other people (Keane, Wolfe, & Taylor, 1987). A brief reminder of the tragic experience can trigger a flashback that borders on panic. Many mundane events become stressful, even years after the original event (Solomon, Mikulincer, & Flum, 1988). In one study, eight Vietnam veterans with PTSD watched a 15-minute videotape of dramatized combat. Watching the film elevated their endorphin levels just as people generally react to an actual injury (Pitman, van der Kolk, Orr, & Greenberg, 1990).

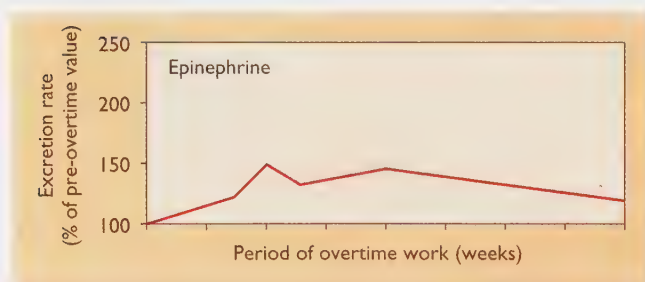


FIGURE 12.24 Epinephrine levels rose for a group of women before, during, and after a long period of working overtime. To establish a baseline, each woman's epinephrine level prior to the overtime period was given as 100%; thus, any number above 100% represents an increase. (From Frankenhaeuser, 1980.)